



Faculty of Computers and Artificial Intelligence

Image Processing (IT 443)

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Hand Gesture Recognition

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Hand Gesture Recognition Using PCA

American Sign Language (ASL) Recognition Project Report

Abstract

This project presents a Hand Gesture Recognition system for American Sign Language (ASL) using Principal Component Analysis (PCA) and Deep Learning techniques. The system aims to recognize 29 ASL categories, including the English alphabet (A–Z) and three special tokens (Space, Delete, and Nothing). The dataset contains approximately 87,000 images collected from an ASL alphabet dataset in the Kaggle environment. The preprocessing pipeline includes image resizing, skin segmentation, edge detection, normalization, and augmentation to improve robustness against lighting and rotation variations. PCA is applied to reduce the dimensionality of image data while preserving 95% of the variance. In addition, a MobileNetV2 transfer learning model is used for classification. The proposed approach improves computational efficiency and supports future integration with real-time webcam inference systems.

Problem Definition and Objectives

Hand gesture recognition is an important application in computer vision and human-computer interaction. Recognizing ASL gestures automatically can help bridge communication gaps between hearing-impaired individuals and others. However, gesture recognition systems face challenges such as background noise, lighting changes, hand orientation differences, and high-dimensional image data.

The main objectives of this project are:

- To preprocess ASL images using resizing, grayscale conversion, skin segmentation, and edge detection.
- To enhance dataset diversity through augmentation techniques such as rotation, zooming, brightness adjustment, and shifting.
- To apply PCA for dimensionality reduction and extraction of the most informative features (EigenHands).
- To train a classification model capable of recognizing 29 ASL gesture categories.
- To prepare the system for future real-time webcam-based gesture recognition.

Methodology

The methodology of the project consists of several stages, including preprocessing, feature extraction, dimensionality reduction, model training, and evaluation. Initially, images were resized to 64×64 pixels to reduce computational complexity and improve PCA efficiency. A preprocessing pipeline was implemented using OpenCV. The pipeline applies skin segmentation in the YCrCb color space, morphological operations to clean the mask, grayscale conversion, and Canny edge detection to highlight hand contours.

To improve model generalization, data augmentation was performed using the Keras ImageDataGenerator. Augmentation techniques included rotation, zooming, width and height shifting, and brightness adjustment. The dataset was divided into 80% training and 20% validation samples.

For feature extraction, each processed image was flattened into a one-dimensional vector. StandardScaler was applied to normalize the data before PCA transformation. PCA retained 95% of the cumulative variance, significantly reducing the number of features while preserving the most important visual information.

In addition to PCA, a deep learning classification model based on MobileNetV2 was implemented using transfer learning. The pretrained ImageNet weights were used as the feature extractor, while custom dense layers, batch normalization, and dropout were added for classification. The model was trained using the Adam optimizer and categorical cross-entropy loss function. Early stopping and learning rate reduction callbacks were applied to prevent overfitting and improve training stability.

Finally, a live inference pipeline was prepared for webcam integration. A temporal smoothing mechanism based on a rolling prediction buffer was implemented to reduce prediction flickering and improve the stability of real-time gesture recognition.

References

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